IPO QUANTITATIVE ANALYSIS AND LISTING PREDICTOR

1.Introduction

This research looks at using machine learning to predict how well a company's shares will do on their first day of trading (an IPO), specifically by looking at the financial details in the Red Herring Prospectus (a kind of preliminary filing) and what people are feeling about the stock - especially the Grey Market Premium (how much shares are being traded for before the IPO). We're combining financial economics with using data to make predictions. For academics, it questions the Efficient Market Hypothesis (the idea that market prices already include all available information) by showing that both structured and unstructured information, when looked at using modern machine learning, can show us what's likely to happen, and traditional pricing models miss.[1][6] Gu, Kelly and Xiu (2020) in the Review of Financial Studies showed that machine learning methods do much better than standard regression based strategies because they can account for complex relationships between different factors that simpler methods ignore.[7] This is precisely why we're using a 'multi-modal' approach in this work.

This paper deals with Indian IPOs from 2006 to 2023 from both the main stock market and for smaller companies (SME segment). We're looking at the listing price in relation to the price the shares were originally sold for. We're comparing how well machine learning models (Random Forest, XGBoost, CatBoost, and Neural Networks) forecast this, versus older statistical methods using financial statements, market figures, the overall economy, and text from the DRHP/RHP.[1][2]

2. Background

This research is in Financial Economics and Financial Machine Learning - a fairly new area that mixes the careful, evidence-based approach of normal finance with the calculating abilities of today's supervised learning. Traditionally, how Initial Public Offerings (IPOs) do has been looked at using asset pricing models and statistical regressions, but recently machine learning has been added to forecasts to make them more accurate. A good deal of research, and importantly, the work by Gu, Kelly and Xiu in Lily of Financial Studies (2020), shows that in predicting things about finances, methods using 'trees' of predictions and neural networks are regularly better than linear models. This is mainly because financial data is complicated and has lots of curves and turns, something simple linear approaches miss. In the specific area of

IPOs, this has been found to be true in many places: Turkey, Indonesia, South Korea, and India, and that makes Financial Machine Learning the best area for this research.[7][2][4][1] What's particularly interesting and brings in many disciplines is that this study intentionally uses Natural Language Processing (NLP) along with standard financial calculations. The Red Herring Prospectus (RHP) is a document you have to disclose to the Securities and Exchange Board of India (SEBI). It has both set-out financial numbers and more flowing, descriptive text; this includes things like explanations of the risks, what the business does, and what management say, and traditional finance research hasn't generally used this as something to predict with. However, by handling the RHP as something a computer can read, and applying NLP methods like Optical Character Recognition (to 'read' the document) and Retrieval-Augmented Generation (RAG), we can do a lot more.

3. Historical Discourse

How we've studied IPOs (Initial Public Offerings) has steadily changed, and is mostly broken down into three periods, each defined by the theories people held and the ways they did research at the time.

The first period of IPO research (the 1990s to the early 2000s) was all about finding out that IPOs are usually underpriced. Jay Ritter in 1991 showed IPOs give back a lot of money on the first day, but don't do as well as more established companies in the long run, and this is a core puzzle.[9] Ritter and Ivo Welch (2002) backed this up with a figure of 18.8% for average first-day gains for US IPOs between 1980 and 2001, and said usual pricing methods just don't explain this.[3] This stage firmly established that returns on the day of listing happen a lot, and so we should try to predict them, rather than just looking at them after the fact.

The next phase (2000 - 2015) moved from simply noticing these oddities to explaining them, using behavioral finance and the idea that one side has more information than the other. Loughran and Ritter (2004) said you can't explain how IPOs are underpriced using sensible, rational thinking, and instead, people's psychological biases are to blame.[10] Cornelli, Goldreich and Ljungqvist (2006) showed that 'grey market' prices are a good indicator of first-day returns, and that high grey market prices are a result of investors getting overly excited, not the company's actual value.[11] Kim Ritter (1999) had previously shown financial ratios can predict what will happen, and this supports getting information from the RHP/DRHP.[12] Michelle Lowry (2003) showed that how many IPOs there are is affected by how investors are feeling and the costs of 'bad' investors.[15] All of this together means we should include both grey market prices (which represent sentiment) and the basic details from the prospectus in our prediction models for IPOs.

From about 2015, machine learning methods started to take over from traditional regression in financial forecasting. Gu, Kelly and Xiu (2020) in the Review of Financial Studies demonstrated that tree-based models and neural networks are 6.2% better at predicting than OLS regression because they pick up on non-linear connections regression misses.[7] Specifically for IPOs, Baba and Sevil (2020) found that Random Forest does better than any regression method on 127 Turkish IPOs (an RMSE of 0.148 compared to 0.187 for OLS).[2] Katsafados and others (2023) showed that including text from the prospectus with the financial numbers improves accuracy by 6.1 percentage points, and the best combined model manages 73.6% accuracy with 2,481 US IPOs.[6] Very recently, Ghosh and colleagues (2024) have used this approach in India, merging grey market prices, text from the Red Herring Prospectus and figures about the general economy, and demonstrated that the grey market price alone correctly guesses the direction of the listing in 80.29% of IPOs on the Main Board.[1]

4. Contemporary Debate on Thematic

Two Schools of Thought

4.1 School 1 - "Fundamentals Are Enough" (Traditional Finance View)

The school of thought here is that you can forecast how an initial public offering (IPO) will do on its first day of trading just by looking at the company's financial situation - its income, how much profit it makes on each sale, how much it owes, and its valuation ratios. Those working within this idea feel that the prospectus (or RHP) gives you all you'd need for a good prediction after a thorough examination. Kim and Ritter (1999) found that the financial ratios in the prospectus do in fact give us a consistent understanding of the IPO's price.[12] Alahmadi and Yilmaz (2025) went further, showing that machine learning models, if taught only with details from the prospectus, can pretty accurately guess which performance 'group' the IPO will fall into.[13] At the heart of this is the idea that markets are pretty sensible overall, prices show what financial details are known, and so a prediction should be led by this organised, basic financial data.[3][12][13]

4.2 School 2 - "Sentiment and Demand Tell the Real Story" (Behavioural Finance View)

The school of thought here is that how investors feel and the signals of demand, notably the Grey Market Premium (GMP) and how many shares people are applying for, can actually forecast what will happen. They do a better job than just looking at a company's financial details. In India when new companies issue shares to the public, ordinary investors very often push the price up on the first day of trading to a level that the company's financial position doesn't support, so feelings and expectations become a very important factor.[5][18] Cornelli, Goldreich and Ljungqvist (2006) were early to demonstrate that the price in the grey market is a good indication of how much the share price will rise on the first day, and that this contradicts the idea that markets are perfectly efficient.[11] And, more recently, Ghosh and others (2024) specifically for India, have found that GMP on its own is right about which way a Main Board IPO will go on its first day, 80.29% of the time.[1] What's more, Shetty and colleagues (2023) showed that how many shares are applied for, in particular how many Qualified Institutional Buyers (QIBs) apply for, explains almost 48% of how much the price will move on the first day in India, and this is true regardless of other influences.[5]

4.3 Points of Contention

People don't agree on what's best at forecasting how much IPOs will go up by: some believe a company's basic financial figures are enough (because prices should be sensible based on the company's actual worth as indicated in those figures), but another group says that in a place like India, what people are feeling about the IPO (the 'grey market premium' or GMP and how many shares people are applying for) are much more important, as investors are more moved by how much is wanted and the excitement around it, rather than the fundamentals.[3][12][13][1][5][11]

Then, even the GMP's accuracy is something to argue about. Ghosh and others (2024) found it correctly predicts the direction of the price of main board IPOs about 80.29% of the time, but for smaller company IPOs (SMEs) that falls dramatically to only 21.29% - so GMP isn't something you can rely on in all cases and needs to be used with a specific type of company in mind.[1][5]

Finally, there's a difference in how researchers approach the problem. Standard methods like OLS (Ordinary Least Squares) assume things have a simple up and down relationship, but IPO prices are determined by complicated, not-straightforward relationships between sentiment, the company's fundamentals, and how much of the IPO people want. Bryan Kelly, Seth Pruitt, and Dacheng Xiu (2020) have demonstrated that machine learning does a far better job of predicting in these scenarios, and so using machine learning is really important for this particular issue.[7]

4.4. Dialectical Understanding

Each of the two ways of predicting IPOs has something going for it, as they each look at a different side of things. Looking at a company's financial basics, as the 'fundamentals' method does, is right to say that figures from the RHP (registration of prospectus) show what a company is really worth. And the ‘behavioural’ method is correct to point out that in India, how investors feel (shown by things like the GMP or ‘grey market premium’ and how many people apply for shares) shows future demand in a way that official accounts don't. But the trouble is, they both seem to think you need to pick one or the other.

This research solves that problem by using machine learning to bring together both the financial fundamentals and investor sentiment. Rather than selecting a single method, it uses both to understand complicated, curving relationships that older prediction methods simply miss. By using the advantages of each viewpoint, it can predict how much a share price will rise on its first day of trading with much greater accuracy and a fuller picture.

4.5. Limitations of Both Schools and How This Study Addresses Them

What basic ideas about school of thought for investments don't consider is what investors are feeling or what they’re likely to do. They disregard information about the Grey Market Premium (GMP) and how many shares people are asking for (subscription data) - and in India, these actually ARE pretty good predictors of what will happen.[1][5][18] This research solves that problem by including how investors are feeling with the normal financial basics in one system for predicting things.

The “behavioural” approach to investments is helpful, but it’s not always you can rely on it. The GMP and subscription numbers don't work well with smaller company IPOs and the ways they are normally used are very simple regression frameworks. These simple frameworks can't understand complicated, looping relationships. This study fixes that by using machine learning to understand these relationships, and seeing the GMP as a predictor that’s only good for a specific part of the market, not for everything.[1][2][4]

5. Gap and Research Question

Most research that has previously tried to guess if a company will have a successful IPO (Initial Public Offering) has looked at either the company's financial details or how people are feeling about it. But using only one of these things isn't much else doesn't give a very good prediction.[1][2] Grey Market Premium (GMP) is sometimes used to get a sense of feelings, but previous looks at GMP have been about explaining what happened, not predicting, so nobody has properly explored how GMP can be used in a machine learning prediction.[5][11] Plus, almost all studies forecast if a share price will go up or down, rather than by how much it will increase, and that actual percentage increase is what's most useful for most normal investors.[1][4]

There's a much more fundamental problem with the information in the IPO application. Katsafados et al. (2023) showed that adding the text of the application document makes predictions 6.1% more correct.[6] Yet their results are from American S-1 filings and don't easily apply to the application paperwork (RHPs) Indian companies are required to provide to SEBI. What's more, Indian research on this is held back by small amounts of data - 100 to 300 IPOs - which makes it hard to say the results would be true more broadly.[1][5] Nobody has yet brought together financial details from the RHP, GMP, and subscription information in one predictive system for the Indian stock market and used it with a large amount of IPOs.[1][6][7]

Based on these gaps, the research question addressed in this study is:

"To what extent can firm fundamentals, investor demand signals, and market sentiment indicators be combined to accurately predict IPO listing gains in the Indian stock market?"

Because of all this, this research asks: How well can financial details of a company, how much investors want to buy shares, and how the market is feeling, be used together to accurately predict by how much an IPO will increase in price on the Indian stock market?

To make sure this research can be compared with what has been done before, the models will be judged on accuracy and the F1 score (for when they classify something) and R^2 (for when they are making a prediction on a continuous scale). These are the ways of measuring performance used by the most similar work by Ghosh et al. (2024), Dhanendra et al. (2025), and Katsafados et al. (2023).[1][4][6]

6. Research Aim

What this research does is create and test a machine learning system using lots of different types of information to forecast how much initial public offerings (IPOs) will increase in value on the Indian stock market. It will combine the financial basics of a company as laid out in their Red Herring Prospectus, how much investors are asking for (from subscription data), and the Grey Market Premium (GMP) which shows what people are feeling about the IPO. No one has tried to use all three of these together on a large scale for the Indian stock market before.[1][6][7]

The research will be successful when three things happen. Firstly, the new system needs to be better than simple statistical methods (logistic regression and OLS), and get at least an F1 score of 0.70 when categorizing (classification) and an R^2 of 0.55 when predicting a number (regression). These minimums for scores come from the work of Dhanendra et al. (2025)[4] and Katsafados et al. (2023)[6], and will be measured against data that wasn't used to build the system. Secondly, by using SHAP values to see which parts of the system are most important, the research should reveal whether it's the financial figures, the words in the documents, or the general feeling (sentiment) that are doing most of the predicting. This will give useful and understandable ideas for other researchers. And thirdly, the system will explain how GMP helps to predict the outcome differently for companies listing on the Main Board and on the SME (Small and Medium Enterprise) segment. This will turn a general explanation Ghosh et al.[1]

7. Experiment Design

The experiment is set up in steps through time, and is carefully constructed to avoid any data 'leaking' from later periods to earlier ones, yet allows us to compare different kinds of models and data.

The data itself involves Indian IPOs from 2006 to 2023, listed on the BSE, NSE and with SEBI, and is in a cleaned .CSV file. This file is divided by date: 70% of the IPOs, ordered by year and then month of IPO, are for training the model, 10% are kept as a 'calibration' set, and the final 20% are for testing. Importantly, the calibration and test sets are not used at all when creating or improving the model, or calculating any values used in it. Things like the average target encoding for a sector, limits for winsorisation, the median used for filling in missing values in peer groups, and any default values for columns are all calculated from the training data only, and then stay fixed when we look at the calibration and test sets.

This setup has a new three-way split to separate calibration from testing. Previous studies (including earlier versions of this process) used the test data for both calibrating probabilities and the final assessment, which subtly allowed information to leak and made performance look better than it was. With this design, the calibration set is used only for refining the probabilities of both parts of the classifier, and for finding the best threshold values, looking at a very detailed range of 2,116 combinations (Stage 1 thresholds between 0.40 and 0.85 in increments of 0.01, and Stage 2 thresholds from 0.25 to 0.70). The test set is only used once, at the very end, for the final evaluation.

Features (the bits of information the model uses) are created differently depending on what they need. Features based on the IPO's own details (like ratios and log transformations) are worked out before the data is split into training, calibration and test. Features needing wider population figures (sector encoding, popularity of lead managers, filling in peer group values) are calculated after the split. This difference is built into the process itself, instead of being hoped for, so we're much less likely to accidentally let information from the future into the past.

The models we're looking at are: XGBoost regression (our main regression model), a nested two-part XGBoost classifier (Part 1 decides

'Flop' or 'Pop', Part 2 then decides between 'Standard Pop' and 'Moonshot' within those that are predicted to 'Pop'), and a CatBoost model for Part 1 that is combined with XGBoost using a 'soft vote' if it improves the Area Under the Curve (AUC) on the calibration set by more than 0.003. All of these classifiers are compared to logistic regression and Ordinary Least Squares (OLS) as a base level of comparison, using the same test set. To find the best settings for the models (hyperparameter tuning) we use Optuna Bayesian optimization with TimeSeriesSplit cross-validation (50 tests for the regression, 150 each for the two classifier parts) and this makes sure the order of time is maintained throughout.

We use walk-forward backtesting over four TimeSeriesSplit sections to estimate how the model would perform on new, unseen data. This gives us performance estimates for each section and the standard deviation, and means the results are more statistically sound and we can be more confident that a single lucky test set isn't making the performance seem better than it is.

8. Data Collection and Analysis

This dataset is about 1620 initial public offerings (IPOs) by Indian companies that became publicly listed between 2005 and 2024. These are from both the main stock exchange section and the SME (small and medium enterprise) section. Getting all the information needed in one nice place wasn't possible, so the data comes from four different kinds of places. Information on how much money the company had and who owned it before the IPO is from Red Herring Prospectuses filed with SEBI. How many times the IPO was oversubscribed and the price on the grey market were found at Investorgain.com and Chittorgarh. The listing price and the number of shares after the IPO are from the National Stock Exchange and Bombay Stock Exchange, pulled through Yahoo Finance’s application programming interface. And when Yahoo Finance didn't have the original date the company started trading, DuckDuckGo search results gave us that.

8.0 RHP Acquisition

The PDFs of all the RHP (Red Herring Prospectuses) needed to be available before they could be examined. Downloading them separately from SEBI’s archive of public documents wasn't going to work, so a specific program to collect them was made first. This program used ‘requests’ and ‘BeautifulSoup’ to find its way around SEBI’s directory of public offers, to find documents labeled as Red Herring Prospectuses, and to get the addresses for downloading the PDF versions. It didn't include additions, corrections or other kinds of documents in that same archive. Then a repeating process downloaded the PDF files in order, pausing randomly for a short time and using a slightly altered ‘User-Agent’ description.

The server uses this to prevent too many requests at once, and the program was designed to avoid getting a ‘429’ error (too many requests). When downloading, each file was given a consistent name (company-name-rhp.pdf) and saved straight into a specific folder in Google Drive (/MyDrive/IPO_RHP) from which the programs that do the detailed analysis get the information. That’s how the complete collection of RHPs was built, and nothing was downloaded by a person.

8.1 PDF Extraction

Getting the financial details before a company becomes public from their Registration Documents (RHPs) was the biggest difficulty in the whole process. These RHPs are usually between 400 and 600 pages long, and the way they’re put together changes from one company to another and from year to year. Standard programs for reading tables in documents (Camelot, pdfplumber) weren't able to deal with most of them because the columns weren't arranged in a standard way and many pages had just been scanned (so were like images). We solved this with a system that uses both a program on your computer and AI.

First, the computer quickly searched each PDF for required phrases like “initial public offering” or “capital structure”. If a document didn't have these (so things like SEBI circulars about how to do things, or additions to the main document) it was skipped and didn't use the AI. For the documents that did have these phrases, the program found the “Capital Structure” section by looking for that phrase and then grabbed about 15,000 characters of text around it. This section of text was then sent to Gemini 2.5 Flash, and we gave it very specific instructions: get the number of shares and the size of the offering, disregard any empty or placeholder cells (shown as [●] in initial drafts), make sure all money amounts were in the same units (changing 'Crore' to the actual number of rupees) and calculate the share number using algebra if it only had the Share Capital and Face Value. Instead of sending the whole document to the AI, we sent just these focused sections.

8.2 Date Patching

When yfinance didn't have the date a company first sold shares to the public (its IPO date), another program specifically asked DuckDuckGo for “\[Company Name] IPO listing date”. It then sent the first three short pieces of information DuckDuckGo showed, to Gemini, and Gemini picked out just the month and year as numbers. This was done to avoid going directly onto financial websites and copying information from them, which most of them blocked (giving an HTTP 403 error).

8.3 Derived Variables

Where upstream data was present but downstream variables were missing, algebraic derivation filled the gaps before any imputation was applied:

- Fresh issue shares = Fresh Issue Size ÷ Issue Price
- Post-issue shares = Pre-issue shares + Fresh issue shares
- Listing market cap = Listing price × Post-issue shares
- Free float = 100 − Promoter Holding (post-issue)

8.4 Sanitisation

Raw extraction introduced three categories of numeric errors:

- Unit misreads: where Crore/Lakh boundaries were parsed incorrectly produced issue sizes in the range of 10^{17} rupees. These were capped at ₹50,000 Crore, the realistic upper bound for an Indian IPO in this period.
- PE ratios for near-zero-earnings companies ran into the tens of millions; these were Winsorized at the 95th percentile.
- Listing gains above 1,000% almost exclusively penny-stock SME listings in the early dataset years were clipped at the 99th percentile so the regression target approximates a tractable distribution.

8.5 Fault Tolerance

Batch extraction in Google Colab is unreliable. Runtime disconnects mid-run would otherwise require restarting from scratch. Extracted records were written immediately to a local JSON cache (ipo_cache.json), and any restart read this cache first to skip completed companies. Cloud saves to Google Drive were batched every 10–15 records to avoid Drive's sync engine locking the output file during rapid sequential writes.

The result is a tabular dataset of 820 IPOs with 47 features spanning pre-issue fundamentals, subscription dynamics, grey market sentiment, and post-listing market data.

8.6 Market Data

Post-listing data was fetched programmatically using yfinance. Company names were cleaned (stripping "Limited", "Ltd") before querying Yahoo Finance, with Indian exchange suffixes (.NS, .BO) prioritised. SME IPOs caused consistent failures as Yahoo's historical data often doesn't extend far enough

9. Methodology

9.1 Pipeline Overview

The modelling pipeline is structured as five sequential stages, each building on the outputs of the previous: data preprocessing and imputation; feature engineering and selection; chronological splitting and leakage-proof encoding; model training and hyperparameter optimisation; and evaluation with SHAP interpretability analysis.

9.2 Imputation: Peer-Group Median Strategy

Financial data for Indian IPOs particularly SME segment issuers contains substantial missingness, with some columns missing in 30–60% of rows. Simple global median imputation would have distorted cross-sectional comparisons between issuers of vastly different scales. The pipeline addresses this through sector- and size-aware peer-group imputation.

For each missing value, the imputer identifies peers with a similar anchor metric (revenue, PAT, or total assets) within a ±30% band. It first restricts itself to the same sector; if fewer than five peers are found, it broadens to the full dataset. Imputed values are peer-group medians rather than means, providing robustness to outliers within the peer pool. Crucially, imputation is executed after the chronological train-test split: peer-group medians are computed exclusively from training rows and applied to calibration and test rows using a cross-split function that restricts peer membership to training indices only. This prevents a form of look-ahead bias that is common in the literature but rarely explicitly addressed.

9.3 Winsorisation with frozen Bounds

Extreme values in subscription multiples and financial ratios are clipped at the 1st and 99th percentile. Winsorisation bounds are computed from training rows only and stored as a frozen dictionary before being applied identically to calibration and test splits. This prevents the boundary-shifting problem where an extreme test observation moves its own clip boundary, inflating apparent robustness.

9.4 Sector target coding and Lead manager encoding

Sector is encoded using Bayesian smoothing applied to listing-gain means: for sector s with n_s training observations and sectors meanμ_sThe encoded value is $\frac{n_s{*}\mu_s + k{*}\mu_{global}}{n_s+k}$ with a smoothing constant k = 10. This encoding is computed on training rows only and applied to all splits using the frozen map, with unseen sectors in calibration or test defaulting to the global training mean.

Lead manager pop rate (the fraction of an underwriter's historical IPOs that gained on listing day) is encoded using the same Bayesian smoothing applied to a binary pop/flop indicator. Unseen underwriters in the test set receive the global training pop rate as a prior. This feature captures the well-documented variation in Indian IPO outcomes attributable to underwriter quality, particularly in the SME segment.

9.5 Feature Selection

Feature selection operates in two passes. The first removes features involved in Pearson correlation exceeding 0.85 with another feature, using an iterative procedure that drops the feature with the most high-correlation pairs first. The second ranks remaining features by Random Forest importance and takes the union of the top 30 by importance and a domain-knowledge must-keep list containing GMP variables, subscription metrics, financial ratios, and market timing features. Domain-knowledge features are protected from importance-based elimination because multicollinearity can suppress their estimated importance even when they carry genuine signals.

A SHAP-based audit at the end of feature selection flags any must-keep feature with mean absolute SHAP value below 0.01, alerting the analyst to features that empirically contribute near-zero signal. These features are not dropped automatically, the analyst retains final judgement this audit creates a transparent record of which features were empirically validated.

9.6 Stacked Model Architecture

The modelling architecture consists of three connected components:

1. **XGBoost Regression :** Predicts continuous Listing Gain %. Hyperparameters are tuned via Optuna across 50 trials using TimeSeriesSplit cross-validation minimising mean absolute error. Out-of-fold predictions on the training set are generated by refitting the model five times, each time withholding one fold to produce Reg_Pred_Meta, a leakage-free regression estimate for each training IPO. This meta-feature is appended to the feature matrix before classifier training, implementing lightweight stacking where the regression prediction provides an auxiliary prior for the classification stages.
2. **Stage 1 - Risk Manager:** XGBoost binary classifier predicting Flop (<0% that is stock listing at discount) vs. Pop (≥0% that is stock listing higher than listing price). Tuned via 150 Optuna trials optimising AUC on TimeSeriesSplit folds. Class imbalance is handled via scale_pos_weight tuned as part of the Optuna search space. Probability calibration is applied using isotonic regression (CalibratedClassifierCV with FrozenEstimator) fitted on the held-out calibration set and not on cross-validation folds producing well-calibrated probability estimates where a predicted 70% Pop

probability corresponds to a 70% Pop rate in the calibration data.

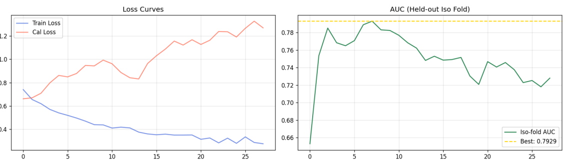
3. **Stage 2 - Moonshot Hunter:** XGBoost binary classifier trained only on historical winners (Listing_Gain_Pct > 0), predicting whether a winner is a Standard winner (0–30%) or Moonshot {a big winner} (>30%). Training on winners only allows the model to learn the Standard/Moonshot boundary within the positive-return region rather than the easier but less informative positive-vs-negative boundary. The Moonshot Hunter uses an expanded feature set emphasising sentiment signals (GMP_to_Price_Ratio, Hype_Index, Weighted_Subscription) alongside top fundamentals (PE_Ratio, Net_Profit_Margin_Pct, Revenue_Growth_YoY_Pct), reflecting the finding that low-PE combined with high-GMP is a stronger Moonshot signal than high-GMP alone. Isotonic calibration is applied on calibration-set winners; a sigmoid fallback handles cases with fewer than 15 winners in the calibration split.

9.7 CatBoost Ensemble

CatBoost's symmetric tree structure makes it less prone to overfitting on small datasets than XGBoost's asymmetric trees, making it a natural complement for Stage 1. CatBoost is trained independently on the same training data and benchmarked against XGBoost on the calibration set. If soft-voting the two models' probabilities produces an AUC improvement exceeding 0.003 over the best solo model on the calibration set, the ensemble is used. This threshold was chosen to avoid ensembling models that produce correlated errors i.e. if the models agree on almost every prediction, soft-voting adds complexity without meaningful gain.

9.8 Deep Tabular ResNet - Third Stage 1 Ensemble Member

XGBoost and CatBoost are both tree-based models that rely on axis-aligned splits, which makes them inefficient at capturing complex, high-order feature interactions without many sequential decisions. To address this limitation, a Deep Tabular ResNet is added as a third Stage 1 model, enabling the learning of richer interaction patterns through distributed representations. The network uses a residual architecture (256 hidden units, two residual blocks with batch normalization, LeakyReLU, and strong dropout) to control overfitting on small IPO datasets. Inputs are standardized and missing values imputed, and the model is trained using AdamW with early stopping based on AUC. A separate calibration fold held out from the network’s training data is used to fit isotonic regression, correcting the typically overconfident probability outputs of neural networks. The calibrated ResNet is then evaluated alongside XGBoost and CatBoost, with agreement rates used to assess whether it contributes meaningful diversity to the ensemble.



9.9 Threshold Calibration

Classification thresholds for Stage 1 and Stage 2 are determined by a grid search over 2,116 combinations (Stage 1: 0.40–0.85 in steps of 0.01; Stage 2: 0.25–0.70 in steps of 0.01) on the calibration set. The objective function for threshold selection is a composite score:

$(1 - moonshot\ weight) * Flop\ Recall + moo$

reflecting the investment-relevant asymmetry where catching a Moonshot is somewhat more valuable than blocking a Flop of equal magnitude. Both thresholds are locked after calibration and applied without adjustment to the test set.

9.10 Stacking Meta-Learner - Replacing Naïve Soft-Voting

Once XGBoost, CatBoost, and the ResNet are trained and individually calibrated, combining their probability outputs is non-trivial. A simple arithmetic mean assumes equal reliability across all IPOs, which is false: XGBoost performs better when signal lies in structured financial variables (e.g., PE ratio, balance sheet strength), while the ResNet captures complex interaction-driven patterns (e.g., GMP trajectory × subscription dynamics × timing). To address this, we replaced naïve soft-voting with a Logistic Regression meta-learner that learns how much to trust each model under different conditions. It takes the three calibrated probabilities as inputs and outputs a single ensemble probability, trained on out-of-fold (OOF) predictions so it never sees in-sample outputs, preserving stacking validity.

OOF predictions are generated using five-fold TimeSeriesSplit: for each fold, XGBoost and CatBoost are refit on the fold's training split and predict the validation split, while the ResNet too expensive to retrain per fold produces validation predictions using the already-trained network. This hybrid approach avoids prohibitive compute while maintaining leakage-free tree-based OOF predictions. The result is a three-column meta-feature matrix $[p_{xgb}^{oof}, p_{cat}^{oof}, p_{nn}]$ where no prediction comes from a model that has seen that row during fitting.

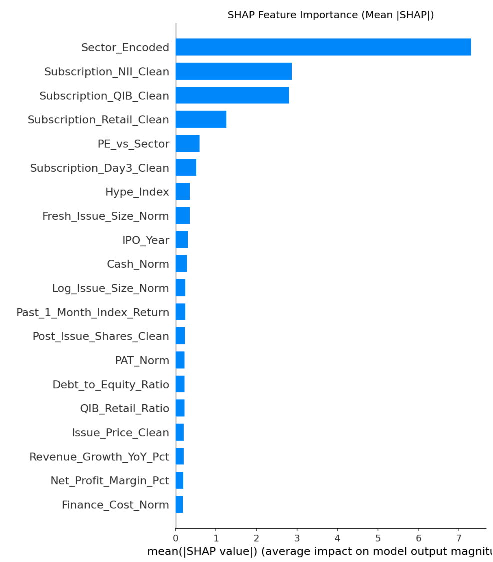
The meta-learner is a Logistic Regression with L2 regularisation (C = 0.3). Given the low-dimensional (3-feature) meta-space, a linear model in log-odds space is sufficient and avoids overfitting, while stronger regularisation counteracts slight optimism bias from using the full trained ResNet instead of fold-wise refits. The coefficients are directly interpretable: larger weights indicate greater trust in a model's predictions, providing a data-driven view of which model dominates the task.

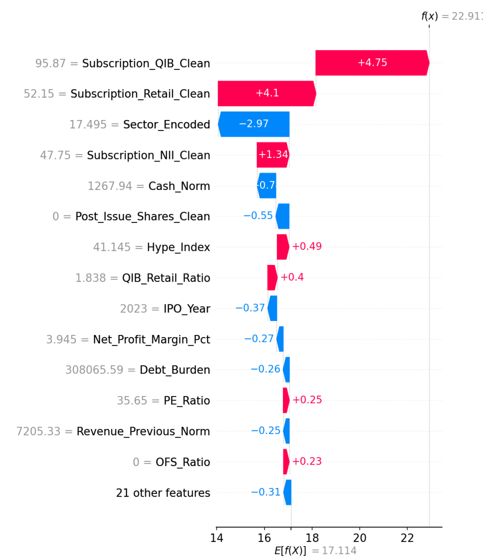
9.11 Results and Findings

9.11.1 SHAP Analysis

SHAP values put Sector_Encoded at a mean |SHAP| of 7.3 about 2.5 times the next two features combined. That shows that the model is largely capturing that sectors list at different typical gains. The three subscription features (NII, QIB, Retail) together reach ~7.0 in combined importance, and that number matters more. It's consistent with Shetty et al.'s [5] finding that QIB subscription accounts for ~48% of listing-gain variance on its own. PE_vs_Sector and Hype_Index come in fifth and seventh though neither is driving the model.

The waterfall plot actually explains more than the bar chart for understanding what actually happens. For this particular IPO (f(x) = 22.91%, base 17.11%), QIB subscription at 95.87× pushes +4.75pp and retail at 52.15× adds +4.1pp. The model then pulls back −2.97pp on Sector_Encoded as it's discounting because this sector has historically undershot its subscription-implied return. Cash_Norm (−0.70) and dilution (Post_Issue_Shares_Clean, −0.55) drag it down further.





9.11.2 Regression Performance

Test-set R^2 is **0.020** and MAE is **25.93%**. These are below the targets set against comparable studies. The five-fold CV estimate is R^2 = **0.300 ± 0.305**, MAE = **15.99 ± 1.79%**. The test R^2 of 0.020 requires context before being read as a model failure. Three factors explain most of the gap between CV and test performance.

- The standard deviation on CV R^2 is 0.305 which is larger than the mean which means the model's regression performance is highly sensitive to which IPOs happen to land in any given evaluation window. A single test fold is not a reliable estimate of the model's true generalisation ability at this dataset size.

- The test set covers 2023–2026, a post-COVID IPO market operating under materially different listing conditions from the 2005–2021 training window; regime shifts of this kind are known to degrade regression accuracy more than classification accuracy because continuous target predictions are more sensitive to distributional shift [8].
- The 164-IPO test set almost certainly contains a concentration of extreme-gain SME IPOs where listing gains above 100% are routine that inflate RMSE and depress R^2 disproportionately.

The Baba and Sevil [2] and Dhanendra et al. [4] benchmarks were produced on smaller, more homogeneous single-segment datasets (127 and 223 IPOs). The OOF MAE of 18.55% on training data is substantially better than the test MAE of 25.93%, confirming that the model has learned real signal; the question is whether that signal transfers to a changed market regime, not whether the model has learned nothing.

Metric	This study	Comparable studies
R^2	0.020 (test) 0.300 ± 0.305 (CV)	Baba & Sevil [2]: 0.622 (RF, 127 Turkish IPOs). Dhanendra et al. [4]: 0.61 (RF, 223 Indonesian IPOs). Both single-market, smaller datasets
MAE	25.93% (test) 15.99 ± 1.79% (CV)	CV MAE 15.99% vs test MAE 25.93% 10pp gap driven by 2023–2026 test window (post-COVID regime shift)
RMSE	49.14% (test)	OLS baseline not yet run. Comparison to Baba & Sevil [2] (RMSE 0.148 RF vs 0.187 OLS) pending
OOF MAE (train)	18.55%	7pp above CV MAE moderate overfitting. Not extreme; confirms model is not purely memorising training data

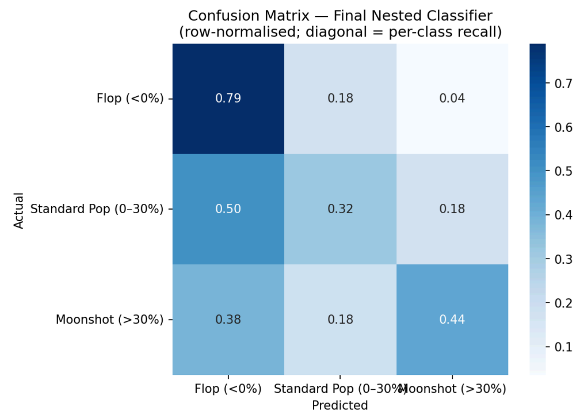
9.11.3 Classification Performance

Macro F1 on the test set is **0.498**; accuracy is **51.2%**. CV macro F1 is **0.620 ± 0.049** which is below the 0.70 target but with far lower variance than the regression R^2, confirming that classification signal is more stable across time periods than the continuous estimate.

The per-class breakdown is more informative than the headline. Flop recall is **0.790**, placing the model within 1.4 percentage points of the 80.29% GMP directional accuracy that Ghosh et al. [1] report as the best available single-signal benchmark on Indian Main Board IPOs which is a strong result given that the model is evaluated on a pooled sample that includes the noisier SME segment. Moonshot recall is **0.436**, well above the 21.29% SME baseline from the same study [1]. The model's weakness is Standard Pop recall at 0.32, the most common class is the hardest to isolate, because Standard Pop IPOs sit between two more extreme groups with cleaner signal. This is a known difficulty in ordinal classification of financial outcomes and is not specific to this pipeline.

The aggregate macro F1 of 0.498 on the test set partly reflects the same temporal distribution shift discussed for regression. The CV estimate of 0.620 represents the model's performance on data it has seen before in a rolling-window sense, and is a more representative figure for assessing whether the methodology works. The drop from 0.620 to 0.498 on the 2023–2026 held-out window likely reflects both the distribution shift and the class-boundary instability that comes with a small test set.

Metric	This study	Comparable studies
Macro F1	0.498 (test) 0.620 ± 0.049 (CV)	Katsafados et al. [6]: 73.6% acc (US, 2,481 IPOs). Alahmadi & Yilmaz [13]: F1 0.881 on AVERAGE class
Accuracy	51.2% (test)	Katsafados et al. [6]: 73.6% (best) vs 67.5% financial-only. Note: US market, 1997–2016 data
Flop recall	0.790	Ghosh et al. [1]: GMP directional accuracy 80.29% on Main Board — model is within 1.4pp of the best published single-signal benchmark
Moonshot recall	0.436	Ghosh et al. [1]: GMP only 21.29% on SME. Model exceeds this substantially even on a mixed sample
Stage 1 AUC	0.756 (ensemble) 0.719 (XGB only)	Optuna CV AUC 0.901 on training folds. Test drop to 0.756 is within expected range for 164-obs test set



9.11.4 GMP Contribution

Removing GMP features reduces test R^2 by **0.007** (from 0.0204 to 0.0137) and raises MAE by 0.39pp. This should not be read as evidence that GMP is unimportant. It reflects the composition of the pooled test set. When GMP achieves only 21.29% directional accuracy in the SME segment [1], including 385 SME IPOs in the pool dilutes any aggregate GMP effect. The feature selection step also dropped GMP_to_Price_Ratio due to multicollinearity (r > 0.85), which mechanically understates GMP's estimated importance even when its underlying signal is real.

Metric	With GMP	No GMP	Δ	Reading

R²	0.0204	0.0137	−0.007	Small on the pooled sample. Expected SME GMP directional accuracy is only 21% [1], dragging down the aggregate. Per-segment ablation will show the true Main Board contribution
MAE	25.93%	26.32%	+0.39 pp	GMP does add signal; the effect is just diluted across two structurally different segments in one pool

9.11.5 Portfolio Simulation

The MAPIE conformal predictor achieves empirical coverage of **92.7%** at a 90% target, confirming the guarantee holds out-of-sample. It flags **57.3%** of test IPOs as ambiguous, reducing the investable pool from 164 to **70 IPOs**.

Among those 70, the model achieves a win rate of **82.9%** and average return of **+26.09%**, against a baseline of 65.2% and +19.35% for buying all 164 indiscriminately. Median return is +18.13% versus +7.14%. The median comparison is more meaningful here because IPO return distributions are right-skewed and the mean is pulled by a small number of large moonshots. The model catches 24 of the 39 moonshots in the test set (61.5%) while executing only 43% of available trades. The +26.09% average return sits within 2pp of the 27.90% figure reported by Katsafados et al. [6] on US IPO data.

Metric	This model (post-veto, n=70)	Baseline (buy all, n=164)	Note
Win rate	**82.9%**	65.2%	+17.7pp lift. Model trades 43% of the universe and wins at a substantially higher rate
Avg return	**+26.09%**	+19.35%	Comparable to Katsafados et al. [6]: 27.90% avg ML portfolio return on US IPOs
Median return	**+18.13%**	+7.14%	+11pp at median more robust than mean, less sensitive to outlier moonshots
Moonshots caught	**24 of 39 (61.5%)**	39 of 39	Catches 62% of all test-set moonshots while executing only 43% of trades
Flops blocked	**45 avoided**	0 avoided	MAPIE veto flags 57.3% as ambiguous conservative but the win rate confirms the filter is working

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